Homework 5

Math 742

2/30/2026

Problem 1 — Consistency (C&B Ch 5.5)

Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Exponential}(\lambda)$, where $\lambda > 0$.

Define the estimator

$$\hat{\lambda}_n = \frac{1}{\bar{X}_n}.$$

a) Show that $\bar{X}_n \xrightarrow{p} 1/\lambda$.

b) Use a continuous mapping argument to prove that $\hat{\lambda}_n$ is a consistent estimator of λ.

c) Explain intuitively why consistency holds even though $1/\bar{X}_n$ is nonlinear.

Problem 2 — CLT and Slutsky (Wasserman Ch 5)

Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Bernoulli}(p)$.

Define $\hat{p} = \bar{X}_n$.

a) Use the Central Limit Theorem to show

$$\sqrt{n}(\hat{p} - p) \xrightarrow{d} N(0, p(1-p)).$$

b) Show that

$$\frac{\sqrt{n}(\hat{p} - p)}{\sqrt{\hat{p}(1-\hat{p})}} \xrightarrow{d} N(0, 1).$$

(Hint: Apply Slutsky's theorem.)

c) Explain why this result justifies the large-sample Wald confidence interval for p.

Problem 3 — Delta Method (Wasserman Ch 9.9)

Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Poisson}(\lambda)$.

We know $\hat{\lambda} = \bar{X}_n$.

We are interested in estimating

$$\theta = \log(\lambda).$$

Define $\hat{\theta} = \log(\bar{X}_n)$.

a) Use the CLT to derive the asymptotic distribution of $\bar{X}_n$.

b) Apply the Delta Method to find the asymptotic distribution of $\hat{\theta}$.

c) Give the approximate asymptotic variance of $\hat{\theta}$.

Problem 4 — Asymptotic Normality of the MLE (Wasserman 9.9)

Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, where σ^2 is known.

a) Derive the MLE $\hat{\mu}$.

b) Compute the Fisher Information for one observation.

c) Use the general MLE asymptotic normality theorem to derive the asymptotic distribution of $\hat{\mu}$.

d) Verify that this matches the exact finite-sample distribution.

Problem 5 — Hypothesis Testing Framework (C&B Ch 8)

Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, where σ^2 is known.

We test:

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_1 : \mu \neq \mu_0.$$

a) Define the Type I and Type II errors in this context.

b) Derive a level-α test.

c) Compute the power function.

d) Show that the power increases with n. If power $\to 1$ as $n \to \infty$, a test is consistent. Is this test consistent?

Homework 9 Math 742 David Cortes

Problem 1

(a) We want to show: for any $\varepsilon > 0$,

$$P(|\bar{X}_n - 1/\lambda| > \varepsilon) \to 0 \text{ as } n \to \infty$$

First compute the mean and variance of $\bar{X}_n$:

$$E[\bar{X}_n] = \frac{1}{n}\sum_{i=1}^{n} E[x_i] = \frac{1}{\lambda}$$

$$Var(\bar{X}_n) = \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{1}{n^2}\cdot\frac{n}{\lambda^2} = \frac{1}{n\lambda^2}$$

Apply Chebyshev's Inequality. Which states that for any random variable Z with finite mean and variance:

$$P(|Z - E[Z]| > \varepsilon) \le \frac{Var(Z)}{\varepsilon^2}$$

Applying this to $\bar{X}_n$:

$$0 \le P(|\bar{X}_n - 1/\lambda| > \varepsilon) \le \frac{Var(\bar{X}_n)}{\varepsilon^2} = \frac{1}{n\lambda^2\varepsilon^2}$$

As $n \to \infty$: $\frac{1}{n\lambda^2\varepsilon^2} \to 0$ Then $P(|\bar{X}_n - 1/\lambda| > \varepsilon) \to 0$.

(b) $\hat{\lambda}_n = \frac{1}{\bar{X}_n}$ is consistent via CMT

Define $g(x) = \frac{1}{x}$. By The CMT If $\bar{X}_n \xrightarrow{P} \frac{1}{\lambda}$ and g is continuos at $\frac{1}{\lambda}$ then $g(\bar{X}_n) \xrightarrow{P} g\left(\frac{1}{\lambda}\right)$

Therefore:

$$\hat{\lambda}_n = \frac{1}{\bar{X}_n} \xrightarrow{P} \frac{1}{1/\lambda} = \lambda$$

(c) Even though $g(x) = \frac{1}{x}$ is nonlinear, it is continuos at $x = \frac{1}{\lambda}$

Continuos means: If the input gets arbitrary close to $\frac{1}{\lambda}$, the output gets arbitrarily close to λ so as $\bar{X}_n \to \frac{1}{\lambda}$, automatically follows that $\frac{1}{\bar{X}_n} \to \lambda$.

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Problem 1. (c)

Nonlinearity causes problems for bias (in finite samples, $E[1/\bar{X}_n] \neq \lambda$), but not for consistency, which only cares about where the estimator converges to - and continuous functions preserve limits.

Problem 2. - CLT and Slutsky

(a) By the CLT $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$

substituting $\bar{X}_n = \hat{p}$, $\mu = p$, and $\sigma^2 = p(1-p)$:

$$\sqrt{n}(\hat{p} - p) \xrightarrow{d} N(0, p(1-p))$$

(b) We want to replace the unknown $p(1-p)$ in the denominator with the estimator $\hat{p}(1-\hat{p})$.

(i) Show the denominator converges in probability.

Define $g(x) = \sqrt{x(1-x)}$, which is continuous on $(0,1)$. Since $\hat{p} \xrightarrow{P} p$ by WLLN and CMT:

$\sqrt{\hat{p}(1-\hat{p})} \xrightarrow{p} \sqrt{p(1-p)}$

(ii) Apply Slutsky's Theorem.

Slutsky states: if $A_n \xrightarrow{d} A$ and $B_n \xrightarrow{P} b$ (a constant), then $\frac{A_n}{B_n} \xrightarrow{d} \frac{A}{b}$

set:

- $A_n = \sqrt{n}(\hat{p} - p) \xrightarrow{d} N(0, p(1-p))$
- $B_n = \sqrt{\hat{p}(1-\hat{p})} \xrightarrow{P} \sqrt{p(1-p)}$

Therefore:

$$\frac{\sqrt{n}(\hat{p} - p)}{\sqrt{\hat{p}(1-\hat{p})}} \xrightarrow{d} \frac{N(0, p(1-p))}{\sqrt{p(1-p)}} = N(0,1)$$

Problem 2. (c)

The result in (b) gives us a pivot — a statistic whose asymptotic distribution $N(0,1)$ does not depend on the unknown p. This is exactly what we need to construct a confidence interval.

For large n, we can write:

$$P\left(-Z_{\alpha/2} \leq \frac{\sqrt{n}\,(\hat{p}-p)}{\sqrt{\hat{p}(1-\hat{p})}} \leq Z_{\alpha/2}\right) \approx 1-\alpha$$

Solving the inequality for p gives the Wald interval:

$$\hat{p} \pm Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

The key steps that justify it are:

1. CLT provides the asymptotic normality of $\hat{p}$
2. Slutsky allows replacing unknow $p(1-p)$ with the consistent estimator $\hat{p}(1-\hat{p})$ without changing the limiting distribution
3. The result is valid only for large n, since both CLT and Slutsky are asymptotic results.

Problem 3. Delta Method

(a) For $X_i \overset{iid}{\sim} \text{Poisson}(\lambda)$: $\longrightarrow$ $E[X_i] = \lambda$, $Var(X_i) = \lambda$

By CLT: $\sqrt{n}\,(\bar{X}_n - \lambda) \xrightarrow{d} N(0, \lambda)$

(b) We apply the Delta Method, which states: If $\sqrt{n}\,(\bar{X}_n-\lambda) \xrightarrow{d} N(0,\sigma^2)$ and g is differentiable at λ with $g'(\lambda) \neq 0$, then:

$$\sqrt{n}\,(g(\bar{X}_n)-g(\lambda)) \xrightarrow{d} N(0, \sigma^2[g'(\lambda)]^2)$$

Set $g(x) = \log(x)$, so:

$$g'(x) = \frac{1}{x} \Rightarrow g'(\lambda) = \frac{1}{\lambda}$$

Applying the Delta Method with $\sigma^2 = \lambda \rightarrow \sqrt{n}\,(\log(\bar{X}_n)-\log(\lambda)) \xrightarrow{d} N(0, \lambda \cdot \frac{1}{\lambda^2}) \rightarrow$

$$\sqrt{n}\,(\hat{\theta}-\theta) \xrightarrow{d} N\left(0, \frac{1}{\lambda}\right)$$

Problem 3. Part (c). Asymptotic variance of $\hat{\theta}$

From (b), the asymptotic variance is: $\text{Asy Var}(\hat{\theta}) = \frac{1}{n\lambda}$

since λ is unknown, we plug in the consistent estimator $\hat{\lambda} = \bar{X}_n$

Giving the estimated asymptotic variance: $\widehat{\text{Asy Var}}(\hat{\theta}) = \frac{1}{n\bar{X}_n}$

Problem 4. Asymptotic Normality of the MLE

Part (a) Derive the MLE $\hat{\mu}$

The likelihood function for $X_1, \dots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ is:

$$L(\mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i-\mu)^2}{2\sigma^2}\right)$$

Taking the log-likelihood:

$$\ell(\mu) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

Taking the derivative and $= 0$ $\quad \frac{d\ell}{d\mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = 0$

$$\sum_{i=1}^{n} X_i - n\mu = 0 \rightarrow \hat{\mu} = \bar{X}_n$$

Part (b) Fisher Information for one observation

Fisher Inf is defined as: $I(\mu) = -E\left[\frac{d^2\ell_1}{d\mu^2}\right]$

$$\ell_1(\mu) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(X_i-\mu)^2}{2\sigma^2}$$

First D: $\frac{d\ell_1}{d\mu} = \frac{X_i-\mu}{\sigma^2}$, Second D: $\frac{d^2\ell_1}{d\mu^2} = -\frac{1}{\sigma^2}$

$$\rightarrow I(\mu) = -E\left[-\frac{1}{\sigma^2}\right] = \frac{1}{\sigma^2}$$

Part (c) Asymptotic distribution of $\hat{\mu}$ via MLE theorem

The general MLE asymptotic normality theorem states: $\sqrt{n}(\hat{\mu}-\mu) \xrightarrow{d} N\left(0, \frac{1}{I(\mu)}\right)$

substituting $I(\mu) = \frac{1}{\sigma^2}$: $\sqrt{n}(\hat{\mu}-\mu) \xrightarrow{d} N(0, \sigma^2)$

Problem 4. Part (d)

Since $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$, the sample mean $\bar{X}_n$ is a sum of independent normals, so we can compute its exact distribution:

$$\bar{X}_n = \frac{1}{n}\sum X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\sqrt{n}(\bar{X}_n - \mu) \sim N(0, \sigma^2)$$

Which matches the Asymptotic result from part (c)

Problem 5. Hypothesis Testing Framework

Let $X_1, \dots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, σ^2 known.

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_1 : \mu \neq \mu_0$$

Part (a): Type I and Type II errors

Error	Def.	In here
Type I	Reject H_0 when H_0 is true	conclude $\mu \neq \mu_0$ when $\mu = \mu_0$
Type II	Fail to reject H_0 when H_1 is true	conclude $\mu = \mu_0$ when $\mu \neq \mu_0$

$$\alpha = P(\text{Type I}) = P(\text{Reject } H_0 \mid \mu = \mu_0)$$

$$\beta(\mu) = P(\text{Type II}) = P(\text{fail to reject } H_0 \mid \mu \neq \mu_0)$$

Part (b): Derive a level-α Test

Test Statistic: Under H_0, since $\bar{X}_n \sim N(\mu_0, \sigma^2/n)$:

$$Z_n = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1) \text{ under } H_0$$

Rejection Region: For a two-sided test at level α:

$$R = \{ |Z_n| > Z_{\alpha/2} \}$$

where $Z_{\alpha/2}$ is the upper $\alpha/2$ quantile of $N(0,1)$

$$P(\text{Reject } H_0 \mid \mu = \mu_0) = P(|Z_n| > Z_{\alpha/2}) = \alpha$$

Part (c): Power function measures the probability of rejecting H_0 for any true value of μ:

$$\pi(\mu) = P(|Z_n| > Z_{\alpha/2} \mid \mu)$$

Under any true μ, the test statistic satisfies:

$$Z_n = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} + \frac{\mu - \mu_0}{\sigma/\sqrt{n}} \sim N(\delta, 1); \text{ where } \delta = \frac{\mu - \mu_0}{\sigma/\sqrt{n}}$$

Problem 5. (c)

therefore the power function is:

$$\pi(\mu) = P(Z > z_{\alpha/2} - \delta) + P(Z < -z_{\alpha/2} - \delta)$$

part (d) Show that the power increases with n. If power $\to 1$ an $n \to \infty$, a test is consistent. Is this test consistent?

Show power increases with n: Fix any $\mu \neq \mu_0$ and let $\Delta = \mu - \mu_0 \neq 0$.

The non-centrality parameter is $\delta = \frac{\sqrt{n}\,\Delta}{\sigma}$

As n increases, $|\delta|$ increases, so:
* $z_{\alpha/2} - \delta \to -\infty \Rightarrow \Phi(z_{\alpha/2} - \delta) \to 0$
* $-z_{\alpha/2} - \delta \to \pm\infty$ depending on sign of Δ

In either case, $\pi(\mu) \to 1$ as $n \to \infty$.

Is this test consistent?

Yes, since for any fixed $\mu \neq \mu_0 \to \pi(\mu) \to 1$ as $n \to \infty$

The test is consistent. It will detect any fixed departure from H_0 with probability tending to 1.